

# SAKSHAM VARSHNEY

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🌐 Portfolio

in sakshamvarshney07

🔗 Saksham-Varshney07

## EDUCATION

**KJ Somaiya School Of Engineering**

2023 – 2027

*Bachelors of Technology in Information Technology - CGPA - 9.06*

## EXPERIENCE

**AI/ML Intern**

Feb 2026 – April 2026

*SkipQ*

- Resolved severe occlusion and lighting failures in retail environments by fine-tuning **YOLOv8** on a **5,000+ image** dataset, achieving a highly robust **89.4% mAP50** accuracy.
- Engineered a real-time inference backend operating at **15 FPS**, seamlessly fusing AI object detection with hardware-accelerated **EAN-13/QR** scanning via the **Open Food Facts API**.
- Developed an interactive dashboard featuring live bounding-box overlays and manual region of interest cropping, automating checkout flows by generating structured **JSON payloads** for payment gateways.

**Contractor**

May 2025 – Dec 2025

*Outlier.ai*

- Executed rigorous **RLHF** pipelines on Large Language Models, systematically ranking outputs to align AI behavior with safety guidelines.
- Streamlined AI output quality by performing detailed factual accuracy checks and providing structured qualitative feedback, achieving a **30% improvement in model coherence**.
- Prepared multi-step **research prompts** to stress-test AI reasoning, utilizing deep failure analysis to heavily optimize **information retrieval** accuracy.

## PROJECTS

**TubeTrail** | *JavaScript, Firebase (Auth/Firestore), Chrome Extension API*



- Built a Chrome extension that transforms YouTube playlists into a structured learning platform by tracking watch progress, managing multiple playlists, and visualizing learning consistency.
- Engineered a **MutationObserver**-based DOM synchronization engine for YouTube's SPA, maintaining **60 FPS** UI updates, alongside **Firebase** cloud synchronization with **0% data loss** across devices.
- Implemented an anti-cheat progress tracker using the **Visibility API** and introduced **365-day heatmaps**, study streaks, and analytics to encourage long-term learning habits.

**ClaimMax** | *Python, Machine Learning, OpenCV*



- Built an automated verification pipeline using **OpenCV** and ML to detect fraudulent ration cards, accelerating manual screening by **80%**.
- Trained custom machine learning models to identify microscopic **authenticity markers** (holograms, QR codes), significantly minimizing **false-positive claims**.
- Deployed a secure **eligibility gatekeeper** logic that actively intercepts invalid document submissions, ensuring **100% risk-free** insurance payouts.

## SKILLS

**Languages :** C++, C, Python , Javascript

**Development :** HTML, CSS, NodeJS, ExpressJS

**Databases :** PostgreSQL, MySQL, MongoDB, Firebase (Firestore), SQL

**AI/ML :** PyTorch, Pandas, NumPy, Machine Learning, Neural Networks

**Others :** DSA, Computer Networks, DBMS, Cybersecurity

**Tools :** Git, Unix